

We did not utilize any LLM in any part of this assignment. All information, ideas, and writing are of our own.

Project Overview

Ultimately, the main objective of this project is to build a web application named JobRec to help job-seekers with finding jobs that best match their skills through aggregated listings from company-specific recruiting sites and a job matching algorithm based on LLM-quantified metrics of relevancy to the user's work experience.

This platform is being developed due to the fact that current job recommendation platforms, such as LinkedIn, and JobRight.ai, often recommend positions which do not properly align with a user's skills, experience level, or career goals. This has resulted in many job applicants wasting time in applying to irrelevant positions, or having to manually look through job positions. JobRec overall seeks to address this problem by allowing users to upload their resumes or input their professional experience into the system, where an AI-based resume analysis tool identifies relevant skills and qualifications. Through extracting these skills, the user's experiences are then compared with aggregated job listings that have been web-scraped from company recruiting sites and other job platforms across the internet to determine the best open job positions based on the user's skills. After parsing the data through this algorithm, it will then provide users with the most accurate and skill-relevant job postings. JobRec will also provide explanations for why these jobs have been recommended.

By improving recommendation accuracy, utilizing user preferences, and giving accurate reasoning as to why certain jobs have been recommended to the user, JobRec hopes to aid job seekers in the current harsh job market by giving transparent, correct reasoning behind job recommendations, and providing a smoother job application process overall.

Scope

In Scope:

Resume upload and skill extraction

Users will be able to upload resumes in a PDF format, after which a skills analysis will be performed, recording their skills for later use.

User accounts and data management

Users will have all relevant information about themselves stored on their account, meaning they will not have to upload their resume again unless they want to update the information. The platform will manage their data securely.

Capability for job extraction

The project will have to pull jobs from some database; the capacity to do so will be within the scope of this project.

Skill-based job recommendations

Users will be recommended jobs that cater most to their skills extracted from their resume. The platform will have a functionality to present the user jobs most relevant to their skills.

Basic geographic filtering

Users will be recommended jobs that cater most to their skills with a preference for locations that are close to their location. The user can either manually enter a location or allow their location to be shared.

Under Evaluation:

Skills database

If time allows, it may be beneficial and cause more relevant job recommendations for a database full of predefined skills that we may use by our model to map to specific jobs.

Out of Scope:

Social media capabilities

JobRec will not be LinkedIn; there will be no capacity for networking or posting. The platform is solely focused on providing good job recommendations rather than allowing users to interact with each other actively.

Assumptions

We are assuming that sufficient job postings will be available across the internet for us to use and recommend for our project.

We are assuming that the users are job seekers who already have a resume and already have some knowledge about how to apply for a job.

Constraints

Since our system relies on external APIs, there is a constraint regarding these APIs. The functionality of these services may constrain how well our project may perform.

There are also constraints regarding data quality. Our proposed web scraper could find job descriptions for jobs that are not offered anymore, or it could provide incomplete job descriptions. If our data is not of quality, that would constrain job recommendations.

Requirements Elicitation & Prioritization

Moscow Method Reflection

Team 1 conducted a MoSCoW method brainstorming session on Thursday, March 5th. We reflected on our stakeholders and discussed what we must have, should have, could have, and won't have in our project. We found discussing each individual section useful; we first looked at what we must have, then when we ran out of concrete ideas for that section, we moved on to what we should have, and so on. We found it was useful to discuss the different prioritization of requirements as it made it clearer what is within our scope and what is not within our scope, allowing us all to be on the same page within our project. We didn't find some of the other brainstorming methods useful, like acting as our stakeholders. Below is a screenshot of our moscow method matrix.



Software Requirements

Functional Requirements:

1. FR-001 - Accurate Job Recommendations

- a. Statement:
 - i. When a user has completed their profile, which includes adding their resume, skills, experience level, etc., the system shall generate and provide a ranked list of accurate recommended jobs whose required skills and qualifications are a close match to the user's given experience and skills data.
- b. Rationale:
 - i. The main goal of this platform is to reduce the amount of wrong top job recommendations, by providing users with job roles which accurately match their qualifications. Other platforms have not been able to provide actual accurate job recommendations yet.
- c. Test Method:
 - i. Create 50 sample users, and fill in their profiles with randomized skills and experience levels. For each of the profiles, determine, through the algorithm, the top 10 recommended jobs to their profiles, and verify that at least 8 of the 10 accurately match the user's experience background.
- d. Supporting Context:
 - i. User profile: Contains stored information of a sample user, which includes skills, education level, work experience, and user's job preferences (What types of jobs they are looking for).
 - ii. Ranked list: Actual jobs recommended based on the user's skills, ordered by their overall skills match and relevance to the user's background.
- e. Tracing Information:
 - i. SYS-001 Accurate Job Recommendations
- f. Priority:
 - i. Must Have

2. FR-002 - Valid Job Applications Links/Redirection

- a. Statement:
 - i. When a user is selecting a job listing from their recommendation list, the system should accurately provide a link that will redirect the user to the correct job application page that is associated with the job posting.
- b. Rationale:
 - i. If there are incorrect and broken links that have been provided as "job recommendations", our platform is not working accurately and is incapable of servicing accurate job recommendations to users.
- c. Test Method:
 - i. Randomly sample 100 stored job postings. When opening the application link, ensure that 99% of the links are not broken, are accurately displayed, and direct the user to the actual job posting and application page.
- d. Supporting Context:

- i. Application link: the URL is directing users to the employer's job application page.
 - ii. Correct page: the page that contains the job listing is accurate and matches the description on the JobRec platform.
- e. Tracing Information:
 - i. SYS-002 Accurate Job Data Collection
- f. Priority:
 - i. Must have

3. FR-003 - Accurate Job Data Scraping

- a. Statement:
 - i. When the system is collecting job postings from job sources across the web, the system shall correctly extract and collect the job title, company name, job location, job description, and application link.
- b. Rationale:
 - i. Accurate job data is required to make sure that there are reliable recommendations and correct job information for users.
- c. Test Method:
 - i. Collect 100 job postings from accurate sources. Through these postings, manually verify the extracted fields. Ensure that at least 95% of the stored fields that the algorithm parsed match the original source data.
- d. Supporting Context:
 - i. Supporting job sources: websites/APIs that have been used by the platform to collect job postings.
 - ii. Extraction: retrieval by algorithm of job information from source.
- e. Tracing Information:
 - i. SYS-002 Accurate Job Data Collection
- f. Priority:
 - i. Must have

4. FR-004 - ChatGPT Hallucination Minimization

- a. Statement:
 - i. When the system generates AI descriptions for job recommendations and skills matches, the system shall generate explanations by utilizing only verified data from the user profile and stored job posting information.
- b. Rationale:
 - i. This is because limiting AI's information on the platform to verified data prevents misleading information and increases user trust.
- c. Test Method:
 - i. Generate 50 AI explanations for recommended jobs. Ensure at least 95% of the explanation statements match the actual user's skills that are stored in the database.
- d. Supporting Context:
 - i. AI explanation: text that describes why a job is recommended to a user
 - ii. Verified data: fields that are within a platform database, such as skills,

experience, and job requirements.

- e. Tracing Information:
 - i. SYS-005 Correct AI Assistance
- f. Priority:
 - i. Must Have

5. FR-005 - Location Preference Matching

- a. Statement:
 - i. When a user specifies location preferences, such as certain geographic regions, or remote/hybrid/on-site work, the system will prioritize job recommendations which accurately match the specific location preferences.
- b. Rationale:
 - i. Location constraints are a large factor in job selection for users. When determining accurate job recommendations, it is important to consider location as well.
- c. Test Method:
 - i. Create 50 test profiles that have listed specific location preferences. Ensure that at least 80% of the job recommendations to each user accurately match the location preferences of the sample user.
- d. Supporting Context:
 - i. Location preference: geographic areas, states, cities, or remote work options that have been selected by the user.
- e. Tracing Information:
 - i. SYS-001 Accurate Job Recommendations
- f. Priority:
 - i. Must Have

6. FR-006 - Skill Gap Detection

- a. Statement:
 - i. When the system determines a recommended job requiring skills that are not present in the user profile, the system shall identify and display the missing skills that are required for the job that the user does not have.
- b. Rationale:
 - i. Insights as to what skill gaps the user has for the job helps the user, for the future, understand the qualifications that they need in order to become more eligible for the roles that they want.
- c. Test Method:
 - i. Select 100 recommended jobs where the user does not have some skills. Ensure that the algorithm identifies and displays the specific missing skills that the job posting asks for, that the user does not have, in at least 90% of the cases.
- d. Supporting Context:
 - i. Skill gap: the difference between skills required by a job posting and the skills that have been listed by the user in their profile.

- e. Tracing Information:
 - i. SYS-001 Accurate Job Recommendations
- f. Priority:
 - i. Should Have

Non-functional Requirements:

1. NFR-001 - Fast Webpage Performance

- a. Statement:
 - i. When a user opens the recommendation dashboard, or any page within the user interface, the system shall load accurate and visible page content within 2 seconds under average operating conditions.
- b. Rationale:
 - i. Faster load times make the platform more user-friendly.
- c. Test Method:
 - i. Conduct 500 page load tests under average/expected system load. Ensure that 95% of the time, the page is loaded and ready within 2 seconds.
- d. Supporting Context:
 - i. Visible page content: the website, subpages, resume upload section, etc. Everything that is displayed to the user.
- e. Tracing Information:
 - i. SYS-004 Reliable Platform
- f. Priority:
 - i. Must Have

2. NFR-002 - Modern and Visually Appealing User Interface (UI)

- a. Statement:
 - i. When a user interacts with the platform interface, the system shall provide a visually consistent, aesthetic, and user-friendly design that has an average usability score that is ≥ 80 on the System Usability Scale (SUS).
- b. Rationale:
 - i. An aesthetic, and easy-to-use interface improves user engagement, and platform understanding.
- c. Test Method:
 - i. Conduct usability testing with 10 users. Collect the survey responses, and verify that the SUS score is on average, greater than or equal to 80.
- d. Supporting Context:
 - i. System Usability Scale (SUS): Standard usability measurement scored from 0-100 to determine how user-friendly a platform is.
- e. Tracing Information:
 - i. SYS-004 Reliable Platform
- f. Priority:
 - i. Should Have

3. NFR-003 - User Data Protection

- a. Statement:
 - i. When a user uses an account to access their skills or job recommendations, or their employment history, full name, address, or other PII, JobRec shall verify that they are the owner of the account before providing any information.
- b. Rationale:
 - i. Because JobRec directly interfaces with the user's employee life through job histories, current employment, and future employment opportunities, it is necessary for it to collect PII on them to find accurate and relevant job listings. Since this information can be used by bad actors to impersonate them or perform doxxing attacks, it is critical that this information only be accessible to the user it belongs to.
- c. Test Method:
 - i. We will use the OWASP ZAP webapp scanner to ensure that authentication flows are secure and prevent unwanted access. We will also use the compliance checker tool Compliance Masonry and manual checklist verification to determine whether the website is compliant with NIST SP 800-63B-4 guidelines for authentication.
- d. Supporting Context:
 - i. PII refers to Personally-Identifying Information, which can be used to link a digital identity to a physical one. NIST SP 800-63B-4 are guidelines for a webserver to establish the identity of users and verify that they own the identity they are trying to authenticate with.
- e. Tracing Information:
 - i. SYS-002 user skill profile
- f. Priority:
 - i. Should Have

4. NFR-004 - Data Storage

- a. Statement:
 - i. While JobRec has a large number of stored job applications and skills, when a user requests job recommendations, JobRec shall continue to provide them in a performant way.
- b. Rationale:
 - i. Though we are only expecting double-digit users, the system will still be scraping many thousands of job applications, and a database of skill names for the LLM to use (if that is how we implement consistent skill name usage) will be considerably large. As such, it is imperative that JobRec maintains performance in this scenario and can efficiently filter through that large repository of job listings to find ones relevant to the user.
- c. Test Method:
 - i. The job database will be filled with 10,000 sample job listings. Then, a new user account will be created with a sample resume, and the user will query for job recommendations. The time for the response to be generated will then be recorded. This process will be repeated over 5 trials to

minimize outliers. The resulting average time should remain under 10 seconds.

- ii. We will also measure the performance of job recommendation querying under scenarios where the database contains 10, 100, 1,000, and 10,000 jobs. The time complexity relative to database entries should then be experimentally determined from this data, and it should be at most linear.

d. Supporting Context:

- i. N/A

e. Tracing Information:

- i. SYS-004 Reliable Platform.

f. Priority:

- i. Should Have

Deliverables

Deliverable: A web and API scraper that can be periodically scheduled with a cron job to aggregate job descriptions and locations from online job boards and company recruiting sites, stores them in a database for further use, then tags each listing by their required skills.

- **Requirements:** FR-003 Accurate Job Data Scraping, FR-001 Accurate Job Recommendations, FR-002 Valid Job Applications Links/Redirection, FR-005 Location Preference Matching, NFR-004 Data Storage

Deliverable: A production-ready, deployable binary for a website where job-seekers can search the job database and be automatically recommended jobs based on their existing skillset and locations, and where employers can post job applications, all running on a publicly-accessible server.

- **Requirements:** FR-001 Accurate Job Recommendations, FR-002 Valid Job Applications Links/Redirection, FR-005 Location Preference Matching, FR-006 Skill Gap Detection, NFR-001 Fast Webpage Performance, NFR-002 Modern and Visually Appealing User Interface, NFR-004 Data Storage

Deliverable: A system accuracy report measuring the following on a sample of 100 job listings and 50 user resumes: the number of correctly-extracted skills by the system, the number of hallucinated skills, the percentage of skill gaps correctly identified in each match, the percentage of functioning links to original listings, the human-verified and qualitative accuracy of matched job listings, and the percentage of job recommendations that match user location preferences. Extra listings may also be included in this sample for certain tests, but each must use a minimum of 100.

- **Requirements:** FR-001 Accurate Job Recommendations, FR-002 Valid Job Applications Links/Redirection, FR-003 Accurate Job Data Scraping, FR-004 ChatGPT Hallucination Minimization, FR-005 Location Preference Matching, FR-006 Skill Gap Detection

Deliverable: A security and compliance audit report demonstrating that the authentication methods and data storage behaviors of the system conform to the NIST SP 800-63B-4 Digital Identity Guidelines. The report will include the results of testing authentication flows with OWASP ZAP, manual checklist verification, and gap analysis via Compliance Masonry.

- **Requirements:** NFR-003 user data protection

Deliverable: A spreadsheet containing the time it took to query for job recommendations over 5 trials when the database contained 10,000 jobs. This spreadsheet should also contain data for job recommendation times when the database has 10, 100, and 1,000 entries so that the time complexity of the operation can be determined.

- **Requirements:** NFR-004 Data Storage

Deliverable: A system usability report containing a usability test with 10 users, 500 page

load tests and their load times, and compliance with WCAG 2.2. The latter will be done using the WAVE tool pointed at our production server.

- **Requirements:** NFR-001 Fast Webpage Performance, NFR-002 Modern and Visually Appealing User Interface

Standards

Web Content Accessibility Guidelines (WCAG) 2.2: <https://www.w3.org/TR/WCAG22/>

- This standard is applicable because our project is a web-based system that job-seekers will use to search jobs, upload information, and review recommendations; hence, the interface needs to be accessible to users with visual, motor, and cognitive disabilities. It will impact our design and code by requiring features like keyboard-navigable UI flows, sufficient color contrast, form labels, accessible error messages, readable font sizes, and more. We will ensure compliance with the standard by running automated accessibility checks during development and manually testing workflows with keyboard-only navigation and screen readers.

Digital Identity Guidelines: Authentication and Authenticator Management:

<https://csrc.nist.gov/pubs/sp/800/63/b/final>

- This standard is applicable because our system has user accounts, logins, and will store sensitive profile data such as resumes, skills, job preferences, visa status, locations, and more. As a result, authentication needs to follow a recognized industry standard; this standard provides information related to defining technical requirements for authenticator assurance levels and focuses on authentication over networks. It will impact our design and code by shaping password handling, MFA support, session/authenticator lifecycle rules, and the way we implement signup, login, password reset, and account recovery. We will ensure compliance with the standard by using Auth0, enforcing strong password and session policies, and documenting authentication requirements in our system design and test cases.

OpenID Connect 1.0: https://openid.net/specs/openid-connect-core-1_0.html

- This standard is applicable because since we're using a third-party identity services tool (Auth0) for secure sign-in, OpenID Connect is the standard identity layer built on top of OAuth 2.0, which Auth0 uses, helping to verify end-user identity and obtaining profile claims. It will impact our design and code by determining how the frontend and backend handle login redirects, ID tokens, user claims, session validation, and integration with Auth0. We will ensure compliance with the standard by using a standards-compliant OpenID Connect provider, validating tokens on the backend, and testing authentication flows against the OpenID connect specification instead of using custom login logic.